CASE STUDY

# REVEL-TRENDS-STORYBOARD

*A case study on brainstorming with LLM council using Swiss tournament strategies*

| ROLE | DOMAIN | TECH | TYPE |
|---|---|---|---|
| Generative AI Engineer | AI / Content Marketing | LLMs, RAG, Vector DB | End-to-End AI Pipeline |

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## Executive Summary

Revel-Trends-Storyboard is an AI-powered platform designed to revolutionize how businesses create promotional video content. As a Generative AI Engineer on this project, the work involved architecting a complete, end-to-end generative AI ecosystem — spanning intelligent context gathering, multi-LLM idea generation, interactive story brainstorming, and AI-assisted video/image generation.

The platform addresses a critical pain point for small and medium businesses: the difficulty of producing creative, platform-optimized promotional video content without a large budget or professional team. Revel-Trends-Storyboard eliminates this barrier by guiding businesses through a structured five-step AI pipeline that transforms basic business inputs into a fully developed, shoot-ready video script — and optionally, AI-generated visual assets.

## Project Overview

| | |
|---|---|
| **Product Name** | REVEL-TRENDS-STORYBOARD |
| **Role** | Data Scientist / Generative AI Engineer |
| **Core Problem** | Businesses struggle to create creative, high-quality promotional videos without large budgets or professional teams. |
| **Solution** | A 5-step Generative AI pipeline from context gathering to AI-powered storyline and video/image generation. |
| **Key Technologies** | Large Language Models (LLMs), Vector Databases, RAG (Retrieval-Augmented Generation), Video/Image Generation APIs |
| **Target Users** | Small to medium businesses promoting on Instagram, TikTok, and similar short-form video platforms. |

## Problem Statement

Creating effective promotional video content for platforms like Instagram and TikTok demands a combination of creative ideation, trend awareness, platform knowledge, and production expertise. For small businesses — especially those operating on limited budgets and with only a mobile phone as their filming device — these demands are often insurmountable.

Key challenges identified include:

- Difficulty generating truly unique and platform-native video concepts tailored to a specific business.
- Inability to benchmark ideas against real-world high-performing content in the same niche.
- Lack of structured storytelling support to convert a raw idea into a production-ready script.
- No accessible, affordable AI-assisted creative pipeline for non-professional video creators.

## Objectives

- Design and implement a conversational context-gathering system to capture comprehensive business intelligence from users.
- Build a multi-LLM ideation pipeline (LLM Council) capable of generating, critiquing, scoring, and shortlisting top video concepts.
- Develop an interactive, iterative story generation and brainstorming system that enables user-driven creative control.
- Integrate an optional AI video/image generation module to produce visual assets aligned with the business story.
- Ensure the entire pipeline is practical for businesses with minimal budgets and only a smartphone for filming.

## System Architecture: The 5-Step AI Pipeline

The REVEL-TRENDS ecosystem is structured as five sequential steps, each building on the output of the previous one. Together, they form a complete generative AI pipeline from raw business data to a shoot-ready promotional video concept.

| | |
|---|---|
| **STEP 1** Context Analysis | A conversational AI chatbot gathers comprehensive business intelligence through 8 structured questions covering business type, platforms, target audience, goals, offerings, challenges, filming device, and additional preferences. |
| **STEP 2** Ideation (LLM Council) | A council of four specialized LLM nodes — Ideator, Critic, Moderator, and Judges — collaboratively generates, refines, scores, and shortlists 4–8 unique promotional video concepts tailored to the business. |
| **STEP 3** Human Refined Ideation | The business selects one idea from the shortlist, or merges elements from multiple ideas. An LLM then refines the hybrid concept into a single, coherent, user-approved idea. |
| **STEP 4** Story Generation & Brainstorming | The refined idea is expanded into a full, scene-by-scene video script. An interactive brainstorming feature continuously offers 4 new creative directions the user can accept and weave into the story. |
| **STEP 5** Video / Image Generation | The finalized story is transformed into an optimized prompt by an LLM, which is then passed to a video or image generation API to produce visual assets aligned with the brand and story. |

# Step 1: Context Analysis — Deep Dive

The system begins by deploying a structured conversational chatbot that engages the business user through 8 carefully designed questions. The goal is to extract a rich, structured business profile that will serve as the foundation for all downstream AI processing.

### The 8 Context Questions

- **Q1.** Business type (and optionally the business name)
- **Q2.** Promotional platforms used (e.g., Instagram, TikTok)
- **Q3.** Target audience demographics and interests
- **Q4.** Short-term and long-term business goals
- **Q5.** Products or services offered by the business
- **Q6.** Key challenges the business is currently facing
- **Q7.** Device available for filming promotional videos
- **Q8.** Any additional context or preferences the business wishes to share

### Sample Context Analysis Output

The following is an illustrative example of the structured JSON profile produced after a context analysis session:

| | |
|---|---|
| **Business Type** | Trekking clothes, shoes and equipment selling brand |
| **Platform** | Instagram, TikTok |
| **Target Audience** | Adventure lovers, trekkers, hikers, and travelers (ages 18–45) |
| **Business Goals** | Build a trusted adventure gear brand, increase sales, attract attention with catchy video content |
| **Device Used** | iPhone 11 (phone only, low budget) |

# Step 2: Ideation — The LLM Council

The Ideation step is the intellectual core of REVEL-TRENDS. It employs a council of four specialized LLM nodes that work in sequence — each with a distinct role — to produce a final shortlist of 4 to 8 high-quality, tailored video ideas.

## Node 1: The Ideator

The Ideator is the creative engine of the council. It receives two inputs: the structured business profile from Step 1 and similar, high-performing content retrieved from the vector database via similarity search. Using both inputs, it generates exactly 8 unique, concise, and visually compelling video concepts — each designed to be immediately actionable and easy to grasp at a glance.

For the above business context the 10 ideas generated by the ideator are:

| Idea | Title | One Line Description | Hook | USP | Niche |
|---|---|---|---|---|---|
| 1 | Storm Armor | Rain pours on the jacket. Water beads up. You stay dry inside. | Rain everywhere, you stay dry. | Waterproof and breathable for Himalayan monsoons. | Adventurers facing heavy rain, monsoon trekkers. |
| 2 | Grip Test | Foot on wet rock. The sole grips. The climb continues. | No slip, all grip. | Best-in-class traction on wet granite and loose gravel. | Hikers and climbers tackling slick trails. |
| 3 | Pack Whisper | Open the pack. Stuff gear. It folds flat. Then fills back up. | One bag, endless space. | Modular, expandable trekking pack with smart compression. | Multi-day trekkers with lots of gear. |
| 4 | Thermal Stretch | Cold dawn. You feel warm. Then you cool down as you move. | Heat when you move, cool when you stop. | Adaptive warmth clothing for fluctuating mountain temperatures. | Winter and shoulder-season trekkers. |
| 5 | Dry Pocket | Phone in waterproof pocket. Cross stream. The | The phone stays dry in river crossings. | A waterproof pocket protects electronics during rain and river crossings. | Solo trekkers and gadget-loving hikers. |

| | | | | | |
|---|---|---|---|---|---|
| | | phone stays dry and bright. | | | |
| 6 | Socks Shield | Old socks gave blisters. New socks shield feet all day. | Blister-free miles ahead. | Anti-friction seams and breathable fibers for long hikes. | Long-distance trekkers and pilgrims. |
| 7 | Pop Tent | Wind blows. Throw the tent. It pops up in seconds. | Set up camp in 15 seconds. | Ultra-fast pop-up tent for solo trekkers. | Solo hikers needing quick shelter. |
| 8 | Trail Glow | Low light. Reflective seams glow. A hiker silhouette shines. | See me on the trail after dark. | Reflective accents for safety on dusk-to-dark hikes. | Night trekkers and early-morning hikers. |

## Node 2: The Critic

The Critic reviews the Ideator's 8 concepts alongside the original business details. Its mandate is to improve: it evaluates each idea for creativity, uniqueness, and business alignment, refining weaker ideas and replacing any that fall short. The output is again 8 ideas — but sharpened, more original, and better aligned with the business context.

The response of the critic in the above ideator’s result is:

Here is a brief critique of the 8 ideas, followed by refined ideas in JSON format.

### Collective flaws observed:

- Overly generic product benefits (waterproof, grip) without memorable storytelling or surprise.
- Hooks are basic and rarely create a strong, repeatable “wow” moment.
- Limited variety in tone and setting; most ideas are focused on single test scenarios.
- Weak differentiation between ideas; several feel similar (durability, weather resistance, setup speed).

- Missing Nepal/Himalayan-specific context and realistic, budget-friendly production angles for a mobile shoot.
- Some ideas rely on features that may seem exaggerated or hard to demonstrate convincingly on a phone, risking credibility.
- Opportunities missed for authentic demonstrations (local trekking scenarios, crowdsourced user moments, micro-stories).

The response of the critic optimizing these flaws are:

| Idea | Title | One Line Description | Hook | USP | Niche |
|---|---|---|---|---|---|
| 1 | Storm Armor | A sudden Himalayan downpour; your jacket beads rain and you march on, dry and comfy. | Can a jacket outsmart a Himalayan monsoon? | Waterproof and breathable design engineered for Nepal's heavy monsoon trails. | Adventurers and trekker groups facing monsoon seasons in Nepal |
| 2 | Grip Test | Wet rock, muddy slope—your sole bites while others slip, finishing the climb with a smile. | No-slip grip, even when the trail is slick. | Best-in-class traction on wet surfaces commonly found on Nepalese trekking routes. | Hikers and climbers tackling slick Nepalese trails |
| 3 | Pack Whisper | Open the pack, stuff gear, it folds flat; then unfolds to swallow a tent, stove, and more. | One bag, endless space—and a hidden seat just for you. | Modular, expandable trekking pack with smart compression and a hidden fold-out camp chair. | Multi-day trekkers with heavy gear in Nepal |
| 4 | Thermal Stretch | Dawn chill on a high ridge: you stay warm as you move, then cool | Warm when you move, cool when you stop—even | Adaptive warmth fabric that regulates temperature across | Winter and shoulder-season trekkers across |

| | | | | | |
|---|---|---|---|---|---|
| | | as you stop and rest. | on chilly Nepal treks. | altitude and wind changes. | Nepal's high passes |
| 5 | Dry Pocket | Cross a stream or sudden rain; phone stays dry and bright inside the pocket. | The phone survives river crossings without a second thought. | Waterproof pocket with touchscreen-friendly window for electronics in rain and river crossings. | Solo trekkers and gadget lovers trekking Nepal’s rivers and monsoon trails |
| 6 | Socks Shield | Old socks blister you; new socks shield feet from dawn to dusk on long days. | Blister-free miles on Nepal’s rugged trails. | Anti-friction seams, breathable fiber and built-in arch support for long treks. | Long-distance trekkers and pilgrims walking Nepal’s routes |
| 7 | Pop Tent | Wind gusts sweep a campsite; you toss the tent and it pops up in seconds with a click. | Set up camp in 15 seconds—solo trekker, no compromise. | Ultra-fast pop-up tent designed for solo trekkers, with secure auto-lock poles. | Solo hikers needing quick, reliable shelter on Nepalese trails |
| 8 | Trail Glow | Low light on a ridge; reflective seams glow softly as you move, guiding your path. | Be seen after dark—safety with a glow-on-the-move. | Reflective accents and glow-reactive stitching for heightened safety from dusk to dawn. | Night trekkers and early-morning hikers across Nepal’s popular routes |

## Node 3: The Moderator

The Moderator performs quality control by cross-referencing the Critic's 8 ideas against a detailed 10-metric evaluation scorecard. If an idea's content does not match its assigned scores, the Moderator revises it until the idea and its scores are consistent. The 10 evaluation metrics are:

| Metric | Description |
|---|---|
| **Originality** | How new, rare, and innovative the idea is vs. typical social media content |
| **Fluency** | How many different variations or executions of the idea can be made |
| **Flexibility** | How adaptable the idea is across niches, creators, or content categories |
| **Feasibility** | How easy or realistic it is to film using just a phone |
| **Practical Value** | How much useful, shareable, or viral value it provides to viewers |
| **Surprise Factor** | How unexpected or attention-grabbing the twist or hook is |
| **Combinatorial Novelty** | How creatively familiar elements are combined in new ways |
| **Scalability** | How many times the idea can be repeated as a series or format |
| **Cultural Freshness** | How well the idea connects to current trends or cultural conversations |
| **Business Alignment** | How well the idea ties to the specific business details provided |

The metric scores provided during moderation are:

| Metric | Score |
|---|---|
| **Originality** | 4.5 |
| **Fluency** | 4.7 |
| **Flexibility** | 5 |
| **Feasibility** | 5 |
| **Practical Value** | 4.3 |
| **Surprise Factor** | 3.8 |
| **Combinatorial Novelty** | 4.2 |
| **Scalability** | 4.1 |
| **Cultural Freshness** | 3.8 |
| **Alignment with business details** | 3.9 |

The response of the moderator to the critic's result in reference to the given scores are:

| Title | One Line Description | Hook | USP | Niche |
|---|---|---|---|---|
| Storm Armor | A sudden Himalayan downpour; your jacket beads rain and you march on, dry and comfy. | Can a jacket outsmart a Himalayan monsoon? | Waterproof and breathable design engineered for Nepal's heavy monsoon trails. | Adventurers and trekker groups facing monsoon seasons in Nepal |
| Grip Test | Wet rock, muddy slope—your sole bites while others slip, finishing the climb with a smile. | No-slip grip, even when the trail is slick. | Best-in-class traction on wet surfaces commonly found on Nepalese trekking routes. | Hikers and climbers tackling slick Nepalese trails |

| | | | | |
|---|---|---|---|---|
| Pack Whisper | Open the pack, stuff gear, it folds flat; then unfolds to swallow a tent, stove, and more. | One bag, endless space—and a hidden seat just for you. | Modular, expandable trekking pack with smart compression and a hidden fold-out camp chair. | Multi-day trekkers with heavy gear in Nepal |
| Thermal Stretch | Dawn chill on a high ridge: you stay warm as you move, then cool as you stop and rest. | Warm when you move, cool when you stop—even on chilly Nepal treks. | Adaptive warmth fabric that regulates temperature across altitude and wind changes. | Winter and shoulder-season trekkers across Nepal's high passes |
| Dry Pocket | Cross a stream or sudden rain; phone stays dry and bright inside the pocket. | The phone survives river crossings without a second thought. | Waterproof pocket with touchscreen-friendly window for electronics in rain and river crossings. | Solo trekkers and gadget lovers trekking Nepal’s rivers and monsoon trails |
| Socks Shield | Old socks blister you; new socks shield feet from dawn to dusk on long days. | Blister-free miles on Nepal’s rugged trails. | Anti-friction seams, breathable fiber and built-in arch support for long treks. | Long-distance trekkers and pilgrims walking Nepal’s routes |
| Pop Tent | Wind gusts sweep a campsite; you toss the tent and it pops up in seconds with a click. | Set up camp in 15 seconds—solo trekker, no compromise. | Ultra-fast pop-up tent designed for solo trekkers, with secure auto-lock poles. | Solo hikers needing quick, reliable shelter on Nepalese trails |
| Trail Glow | Low light on a ridge; reflective seams glow softly as you move, guiding your path. | Be seen after dark—safety with a glow-on-the-move. | Reflective accents and glow-reactive stitching for heightened safety from dusk to dawn. | Night trekkers and early-morning hikers across Nepal’s popular routes |

## Node 4: The Judges

Two independent Judge LLMs each select their top 4 ideas from the Moderator's refined shortlist. The final output to the user consists of all ideas chosen by either judge. Ideas selected by both judges are elevated as top ideas. This dual-judge mechanism ensures diversity and consensus simultaneously, delivering between 4 and 8 final ideas to the business.

The ideas selected by judge 1 are:

| Title | One Line Description | Hook | Unique Selling Proposition | Niche |
|---|---|---|---|---|
| Pack Whisper | Open the pack, stuff gear, it folds flat; then unfolds to fit a tent, stove, and more. | One bag, endless space—and a hidden seat just for you. | Modular, expandable trekking pack with smart compression and a hidden fold-out camp chair. | Multi-day trekkers with heavy gear in Nepal |
| Pop Tent | Wind blows at the campsite; you pop the tent up in seconds with a click. | Set up camp in 15 seconds—solo trekker, no compromise. | Very fast pop-up tent for solo trekkers, with strong auto-lock poles. | Solo hikers who need quick, reliable shelter on Nepal's trails |
| Trail Glow | In low light on a ridge, reflective seams glow softly as you move, guiding your path. | Be seen after dark—glow while you move. | Reflective details and glow-reactive stitching for safety from dusk to dawn. | Night trekkers and early risers on Nepal’s popular trails |
| Grip Test | Wet rock, muddy slope—your shoe grips while others slip, finishing the climb with a smile. | No-slip grip, even when the trail is slick. | Top traction on wet surfaces found on Nepal's trekking trails. | Hikers and climbers tackling slick trails in Nepal |

The ideas selected by judge 2 are:

| Title | One Line Description | Hook | Unique Selling Proposition | Niche |
|---|---|---|---|---|
| Pack Whisper | Open the pack, stuff gear, it folds flat; then unfolds to fit a tent, stove, and more. | One bag, endless space—and a hidden seat just for you. | Modular, expandable trekking pack with smart compression and a hidden fold-out camp chair. | Multi-day trekkers with heavy gear in Nepal |
| Pop Tent | Wind blows at the campsite; you pop the tent up in seconds with a click. | Set up camp in 15 seconds—solo trekker, no compromise. | Very fast pop-up tent for solo trekkers, with strong auto-lock poles. | Solo hikers who need quick, reliable shelter on Nepal's trails |
| Dry Pocket | Cross a stream or sudden rain; your phone stays dry and easy to read inside the pocket. | The phone stays dry during river crossings. | Waterproof pocket with a window that works with touchscreens. | Solo trekkers and gadget lovers on Nepal's rivers and monsoon trails |
| Grip Test | Wet rock, muddy slope—your shoe grips while others slip, finishing the climb with a smile. | No-slip grip, even when the trail is slick. | Top traction on wet surfaces found on Nepal's trekking trails. | Hikers and climbers tackling slick trails in Nepal |

The final selected unique ideas from two judges are:

| Title | One Line Description | Hook | Unique Selling Proposition | Niche |
|---|---|---|---|---|
| Pack Whisper | Open the pack, stuff gear, it folds flat; then unfolds to fit a tent, stove, and more. | One bag, endless space—and a hidden seat just for you. | Modular, expandable trekking pack with smart compression and a hidden fold-out camp chair. | Multi-day trekkers with heavy gear in Nepal |
| Pop Tent | Wind blows at the campsite; you pop the tent up in seconds with a click. | Set up camp in 15 seconds—solo trekker, no compromise. | Very fast pop-up tent for solo trekkers, with strong auto-lock poles. | Solo hikers who need quick, reliable shelter on Nepal's trails |
| Trail Glow | In low light on a ridge, reflective seams glow softly as you move, guiding your path. | Be seen after dark—glow while you move. | Reflective details and glow-reactive stitching for safety from dusk to dawn. | Night trekkers and early risers on Nepal’s popular trails |
| Grip Test | Wet rock, muddy slope—your shoe grips while others slip, finishing the climb with a smile. | No-slip grip, even when the trail is slick. | Top traction on wet surfaces found on Nepal's trekking trails. | Hikers and climbers tackling slick trails in Nepal |
| Dry Pocket | Cross a stream or sudden rain; your phone stays dry and easy to read inside the pocket. | The phone stays dry during river crossings. | Waterproof pocket with a window that works with touchscreens. | Solo trekkers and gadget lovers on Nepal's rivers and monsoon trails |

## Step 3: Human Refined Ideation

After receiving the shortlisted ideas, the business takes control. In this step, the user may select a single idea to develop further, or — leveraging the platform's LLM-powered merge capability — combine conceptual elements from two different ideas into one cohesive hybrid concept.

This human-in-the-loop design ensures the final creative direction is always grounded in the business owner's vision. The LLM processes the user's merge instructions and produces a single, refined, ready-to-develop idea that serves as the input for story generation.

Let’s assume this prompt is given by the user:
Combine the ideas of a pop tent and dry pocket and give me a new idea.

Then the response with final ideation is:

| Field | Content |
|---|---|
| title | Pop Shelter |
| one_line_description | Storm hits while trekking; your tent pops up instantly and your gadgets stay dry inside a touch-friendly pocket. |
| hook | Set up camp and protect your tech in seconds—rain or river, solo trek ready. |
| unique_selling_proposition | Ultra-fast pop-up tent with built-in waterproof tech pocket that keeps your phone visible and usable. |
| niche | Solo hikers in Nepal who need instant shelter and gadget protection on unpredictable trails |

# Step 4: Story Generation with Brainstorming

The human-refined idea is now expanded into a complete, scene-by-scene video script. This is not a rough outline — it is a fully detailed, production-ready storyline that a business can follow with their phone camera to shoot a compelling promotional video.

## The Brainstorming Feature

A standout feature of this step is the interactive brainstorming system. Immediately after the initial story is generated, the platform proposes 4 new, diverse creative directions that could deepen or diversify the narrative. These brainstorming suggestions are always fresh — they never repeat scenes or themes already present in the story.

The user can accept one or more brainstorming ideas, and the story automatically regenerates to incorporate the selected directions. This cycle repeats: each new story version spawns 4 fresh brainstorming ideas, allowing the business to iteratively sculpt their story until they are fully satisfied with the result.

- The brainstorming loop is unlimited — users can iterate as many times as needed.
- Each iteration always generates 4 genuinely new creative suggestions.
- Multiple brainstorming directions can be accepted simultaneously.
- The final story is the definitive shoot-ready script for the business.

### Initial story from the above idea:

**Concept title:** Pop Shelter – Instant Shelter, Protected Tech, Anywhere

### Scene 1 — Hook: Storm on the Trail

- Visual: A lone hiker (young adult, Nepal landscape) trudges through rain and wind. Slow-motion raindrops, muddy paths, distant thunder.
- Action: The hiker reaches a compact pack, pulls a single strap, and the Pop Shelter pops up inches from the ground in an instant.
- Audio: Rains, wind, a brief, bold whoosh as the tent deploys; subtle heartbeat-like music builds.
- On-screen text: “Set up camp and protect your tech in seconds—rain or river, solo trek ready.”
- Brand tie-in: Introduces the hero product (instant pop-up tent) as the solution to unpredictable weather.

### Scene 2 — The Problem Before the Shelter

- Visual: Close-ups of the harsh environment: mud-slick trail, a drenched map, a phone screen getting splashed.
- Action: The hiker pulls out gear (durable trekking clothes, rugged shoes, small backpack) from a weatherproof bag; quick cuts show the need for reliable gear in rugged Nepal trails.
- Audio: Subtle narration or on-screen caption: “Durability you can trust on every mile.”
- Brand tie-in: Sets up the brand promise of durable, reliable trekking gear.

### Scene 3 — The Pop Shelter Reveal

- Visual: Ultra-close shot of the tent’s mechanism as it pops open; fabric unfurls rapidly; ground pegs slide in with minimal effort.
- Action: Inside the tent, one corner features a built-in waterproof tech pocket. The phone (iPhone 11) slides into the pocket; the screen remains visible and touch-responsive through a clear waterproof layer.
- Audio: Crisp mechanical sounds of the pop-up action; rain still tapping the exterior.
- On-screen text: “Ultra-fast pop-up. Built-in waterproof tech pocket. Your phone, visible and usable.”
- Brand tie-in: Emphasizes the unique selling proposition directly.

### Scene 4 — Tech Pocket in Action

- Visual: The phone is used to check a map, snap photos, and send a quick message, all while inside the dry tent.
- Action: Close-ups of fingers swiping through the phone, waterproof pocket keeping moisture out.
- Audio: Simple, upbeat tempo; light tapping on a touchscreen.
- On-screen text: “Touch-friendly pocket. Stay connected even in the storm.”
- Brand tie-in: Demonstrates practical value of the tech pocket component.

### Scene 5 — Durability in Real Conditions

- Visual: The hiker steps outside the tent under rain; wind tests the outer fabric, rain beads slide off, gear stays dry inside the tent and in the pack.
- Action: A quick, energetic montage of the tent enduring wind and rain while the hiker continues trekking.
- Audio: Wetter weather ambience mixing with the resolve of the trekker’s pace.
- On-screen text: “Engineered for harsh trails. Dry inside. Ready outside.”
- Brand tie-in: Reinforces durability and protective design.

### Scene 6 — The Gear Ensemble (Brand Promise Spotlight)

- Visual: A fast-cut montage of the rest of the brand’s offerings: durable trekking clothes, rugged shoes, backpack, and a compact tent (other essential equipment).

- Action: The hiker models the clothes, clips the backpack, laces up sturdy shoes, and cinches the tent bag—all via quick, satisfying motions.
- Audio: Energetic, optimistic music with a snappy rhythm.
- On-screen text: “Durable gear for adventure lovers, trekkers, hikers, and travelers.”
- Brand tie-in: Positions the brand as a complete, trusted adventure gear line.

### Scene 7 — Solo Nepal Trekking Moment

- Visual: A wide shot of a sweeping Nepali landscape at golden hour; the hiker stands at a ridge, sun glow, distant yak bells, snow-capped peaks in view.
- Action: The hiker slips the phone from the tech pocket to capture a quick selfie with the tent in the background, then tucks it back as the wind picks up.
- Audio: Ambient nature sounds with a subtle, inspirational score.
- On-screen text: “For solo trekkers who crave reliability on unpredictable trails.”
- Brand tie-in: Deepens connection to the target niche and the brand’s mission.

### Scene 8 — Influencer-Style Inspiration Moment (Creativity within Constraints)

- Visual: A short, stylish montage inspired by popular mobile-style videos: “pack with me” vibe, rapid cuts, natural light, on-screen captions.
- Action: The hiker does a quick “pack-down” visual: clothes, shoes, pack, and tent all stowed efficiently; the Pop Shelter remains the centerpiece.
- Audio: Snappy, punchy cuts with light voiceover: “You can do it too.” (inspired by influencer aesthetics; not a direct promo)
- Brand tie-in: Demonstrates how easy it is to organize and carry gear, aligning with the social-media-friendly nature of Instagram/TikTok content.

### Scene 9 — Call to Action and Endnote

- Visual: The hiker sits at sunset inside the tent, the city lights miles away, phone resting in the tech pocket, ready for a message home or navigation.
- Action: Final push-in on the tent’s label and the brand logo on gear; screen shows social handles.
- Audio: Warm closing note; faint wind.
- On-screen text: “Follow, like, and shop our adventure-ready gear. Link in bio.”
- Brand tie-in: Clear, concise CTA aligned with Instagram and TikTok platforms

**The brainstorming topics from the above story are:**

| Topic | Content |
|---|---|
| topic1 | A Nepali guide's POV follows the hiker, showing how the gear keeps clients safe on a storm-lashed pass, with authentic, culturally textured visuals and on-screen tips from the guide. |
| topic2 | A single gear piece (tent, jacket, or shoes) is tested across four terrains—rain, mud, snow, heat—in a 60-minute, split-screen montage that highlights dryness, grip, and comfort. |
| topic3 | Urban-to-trail crossover: a day-in-the-life reel that demonstrates the same jacket, shoes, and pack being used from city commutes to trailheads, emphasizing everyday practicality and versatility. |
| topic4 | Stop-motion unboxing and transformation: a playful 10–15 second loop showing the gear unfolding from packaging to ready-for-adventure, focusing on tactile motion and the tent's pop-up mechanism. |

Now, the user can include any of these brainstorming topics, with multiple options allowed to give a new direction or dynamics to the story in the infinite loop.

## Step 5: Video / Image Generation

In the final step, the fully brainstormed story is passed to an LLM that generates an optimized prompt for a video or image generation API. The business details from the context analysis are also provided at this stage, ensuring the generated visuals are on-brand, tonally appropriate, and contextually accurate.

This step is intentionally optional. If the business prefers to shoot their own video following the script, the story from Step 4 is sufficient. However, for businesses that want AI-generated promotional assets — for mockups, social media hooks, or inspiration — this step delivers ready-to-use images or short video clips aligned with their story and brand identity.

The sample of generated prompt is:

An ultra-high-resolution promotional still that embodies Pop Shelter, the ultra-fast pop-up tent with a built-in waterproof tech pocket, built for solo trekkers on Nepal's wild trails. In the frame, a lone young adult hiker battles rain on a rain-slick ridge, late-afternoon light breaking through storm clouds to cast a warm glow on distant snow-capped peaks. A compact backpack sits at the hiker's feet as the Pop Shelter snaps open inches from the earth, fabric unfurling in a rapid, almost magical motion while ground pegs slide home with a single maneuver and rain beads shimmer on the outer fabric. In a corner, the tent reveals a built-in waterproof tech pocket holding an iPhone 11; the screen is clearly visible through a transparent, touch-responsive layer, while the hiker's gloved hand reaches toward the pocket with a calm, purposeful gesture. The mood is rugged, determined, and optimistic; the palette mixes earthy greens and stone greys with the tent's warm olive tones and subtle neon accents. The composition feels cinematic, designed for Instagram and TikTok thumbnails, with a concise, brand-forward hint in the tent silhouette and a tag-line about ultra-fast setup and tech protection. Shoot with phone-friendly realism, ready for mobile audiences.

Then this prompt is passed to a video generation model to create the video. We can pass the whole storyline too for better response if the high token usage is not the key consideration.

## Technical Highlights

- Vector Database & RAG: Business context is matched against a database of real promotional video captions — including engagement metrics (likes, comments) — to ground idea generation in proven, high-performing content strategies.

- LLM Council Architecture: A sequential multi-agent pipeline (Ideator → Critic → Moderator → Judges) ensures ideas are generated, challenged, evaluated against 10 metrics, and consensus-filtered before reaching the user.

- Evaluation-Driven Moderation: The Moderator node's use of 10 scoring dimensions (originality, feasibility, cultural freshness, etc.) brings structured quality control typically absent in pure LLM generation pipelines.

- Human-in-the-Loop Design: The pipeline is designed for co-creation — user interventions at Steps 3 and 4 ensure outputs remain grounded in the business owner's intent rather than being fully autonomous.

- Iterative Story Brainstorming: The unlimited brainstorming loop provides a genuinely interactive creative experience, transforming one-shot generation into a dynamic, user-driven narrative sculpting process.

## Key Outcomes & Value Delivered

REVEL-TRENDS successfully delivers a comprehensive generative AI ecosystem that transforms a simple business description into a fully developed, shoot-ready promotional video concept — and optionally, AI-generated visual assets. The key value delivered includes:

- End-to-end automation of the creative ideation-to-script pipeline, reducing creative production time significantly.
- Democratized access to professional-quality video concept development for small businesses with limited budgets.
- Improved idea quality through multi-LLM validation, scoring, and consensus — reducing reliance on any single model's output.
- Increased creative control for business owners through human-in-the-loop intervention points and iterative brainstorming.
- Feasibility-first design throughout — all outputs are calibrated to be executable with just a smartphone.

## Lessons Learned

- Multi-agent LLM pipelines significantly outperform single-model approaches for creative tasks requiring both generation and critical evaluation.
- Injecting real-world engagement data (via vector similarity search) dramatically improves the relevance and virality-potential of generated ideas.
- Evaluation-driven generation — where ideas are scored and then revised to match their scores — is a powerful quality-control mechanism that reduces vague or generic outputs.
- Human-in-the-loop checkpoints are essential for creative AI products where the end user's vision must be preserved and respected.
- Iterative brainstorming (rather than one-shot generation) produces meaningfully richer and more satisfying storylines.

## Conclusion

REVEL-TRENDS represents a significant step forward in applied generative AI for content marketing. By architecting a five-step pipeline that combines conversational intelligence, retrieval-augmented generation, multi-agent LLM councils, human-in-the-loop design, and optional AI visual generation, the project demonstrates how complex creative workflows can be effectively automated while preserving — and even enhancing — human creative agency.

The platform is especially impactful for small and medium businesses that lack the resources of large marketing teams, offering them a structured, AI-guided path from business context to a complete promotional video strategy — all executable with nothing more than a smartphone.

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